

Assignment 5, Part 1
Chelsea Brackett

Suppose we observe $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, where x_i 's are some fixed, non-random numbers with $\min\{x_i\} < \max\{x_i\}$. Assume there exists the following linear relation:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma_i^2) \text{ are independent, } i \in \{1, 2, \dots, n\},$$

with σ_i^2 being dependent on index i (i.e. *heteroscedasticity*).

Further, assume we know that $\sigma_i = w_i \sigma$ for some known $w_i > 0$ and unknown σ .

Because the variances are not equal, minimization of the sum of squared residuals may result in an inefficient estimator: intuitively we understand that observations with large variability should not influence the estimates of β_0 and β_1 in the same way as observations with small variability. Let's then instead rewrite the problem as the following Multivariate Linear Regression model:

$$\underbrace{w_i^{-1} y_i}_{\text{response}} = \beta_0 \cdot \underbrace{w_i^{-1}}_{\text{1st regressor}} + \beta_1 \cdot \underbrace{w_i^{-1} x_i}_{\text{2nd regressor}} + \tilde{u}_i,$$

where $\tilde{u}_i = w_i^{-1} \varepsilon_i$.

- Show that this new model with response $\tilde{y}_i = w_i^{-1} y_i$ and inputs $(w_i^{-1}, w_i^{-1} x_i)$ is homoskedastic. Find $\text{Var}(\tilde{u}_i)$.
- Find the OLS estimators of β_0 and β_1 using this homoskedastic model. Are these estimators unbiased?
- Show that these estimators can be equivalently obtained by minimizing the *weighted least squares* (WLS): $\sum_{i=1}^n w_i^{-2} (y_i - \beta_0 - \beta_1 x_i)^2$.

SOLUTION:

a) The variance of the original model is heteroskedastic:

$$\text{var}(y_i) = \sigma_i^2$$

but in the new model:

$$\text{var}(\tilde{y}_i) = \text{var}(\tilde{u}_i)$$

and if $\tilde{u}_i = w_i^{-1} \varepsilon_i$, then:

$$\text{var}(\tilde{u}_i) = \text{var}(w_i^{-1} \varepsilon_i) =$$

$$\text{var}(\tilde{u}_i) = w_i^{-2} \cdot \text{var}(\varepsilon_i)$$

$$\text{var}(\tilde{u}_i) = w_i^{-2} \cdot (w_i \sigma)^2$$

$$\text{var}(\tilde{u}_i) = \frac{1}{w_i^2} \cdot w_i^2 \cdot \sigma^2$$

$$\text{var}(\tilde{u}_i) = \sigma^2$$

Here σ^2 is not dependent on i so the variance of the new model is constant, which means it is homoskedastic.

normal:

$$b) \hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n w_i^{-2} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n w_i^{-2} (x_i - \bar{x})^2} \quad \hat{\beta}_0 = \left(\frac{\sum_{i=1}^n w_i^{-2} y_i}{\sum_{i=1}^n w_i^{-2}} \right) - \hat{\beta}_1 \left(\frac{\sum_{i=1}^n w_i^{-2} x_i}{\sum_{i=1}^n w_i^{-2}} \right)$$

Yes, the estimators are unbiased because the new, weighted model is homoskedastic and BLUE. Since it's homoskedastic the $E[w_i^{-1} \varepsilon_i] = 0$, so you're left with just $E[\hat{\beta}_0] = \beta_0$ and $E[\hat{\beta}_1] = \beta_1$, which is unbiased.

$$c) \tilde{y}_i = w_i^{-1} y_i = \frac{y_i}{w_i} \quad \tilde{x}_{i1} = \frac{1}{w_i} \quad \tilde{x}_{i2} = w_i^{-1} x_i = \frac{x_i}{w_i} \quad \tilde{u}_i = w_i^{-1} \varepsilon_i = \frac{\varepsilon_i}{w_i}$$

new model

OLS objective

$$\sum_{i=1}^n (\tilde{y}_i - \beta_0 \tilde{x}_{i1} - \beta_1 \tilde{x}_{i2})^2$$

$$= \sum_{i=1}^n \left(\frac{y_i}{w_i} - \beta_0 \frac{1}{w_i} - \beta_1 \frac{x_i}{w_i} \right)^2$$

$$= \sum_{i=1}^n \left(\frac{y_i - \beta_0 - \beta_1 x_i}{w_i} \right)^2$$

$$= \sum_{i=1}^n w_i^{-2} (y_i - \beta_0 - \beta_1 x_i)^2$$

Same as WLS